

Seminar 12

1. Self-Supervised Learning

Supervised deep learning has obtained excellent results in sectors such as computer vision and natural language processing; however, all these models require a large amount of labelled data, the collection of which is expensive, slow, and requires human supervision. To overcome this issue, a different approach was developed: Self-Supervised Learning, which learns from a large amount of unlabelled data and automatically generates pseudo-labels (Gui et al., 2015).

Self-Supervised Learning offers numerous advantages and disadvantages.

The main advantages of SSL are the reduction in the dependence on the labelled data, improved model performance, reduction in the training time, and improved generalisation capability. Although there are some disadvantages, such as high computational costs because SSL algorithms use complex pretext tasks and data augmentation techniques, which require more computational resources than other methods, pseudo-labels can contain errors, leading the model to learn the wrong representation, there is a lack of effective pretext tasks, and as pseudo-labels are not human-supervised, the signal can be inaccurate (Zhao et al., 2024).

2. Advantages and Challenges of Neural Architecture Search

A neural architecture search was created to automatically find the best neural network architecture for a specific dataset without human assistance.

The NAS is based on three fundamental components: a search space that defines all the possible architectures that a system can explore, including the operations, layers, and interconnections among neurons; a search strategy that establishes how to explore this space to identify the best architecture with the minimum number of attempts; and an evaluation strategy that evaluates the architecture's performance and guides the search process to more promising solutions (Wang & Zhu, 2024).

The main advantages of using Neural Architecture Search are as follows: automation of neural network projects, reducing the dependence on developers' expertise; optimizing models for specific hardware such as smartphones, edge devices, and embedded systems; and reducing search time with the introduction of validation strategies that allow for quickly estimating the network performance without complete training.

Neural Architecture Search also has several challenges, such as defining the search space, as a larger search space increases the probability of finding better architectures but requires longer search times; a continuous trade-off between the model's quality and the research costs; and greater computational resources because a NAS requires significant computational capability (Salmani Pour Avval et al., 2025).

3. How Edge AI is enabling AI at the edge

Edge AI moves artificial intelligence processing from the cloud to edge servers, reducing the time for data processing and allowing real-time decision-making.

To make this possible, Edge AI uses different architectures: some process the data on the device completely, some send the data to another edge server, and others distribute the work among the device, edge, and cloud to obtain a balance between performance and resource consumption. This system offers several advantages, such as reducing latency, decreasing network traffic and energy consumption, improving privacy, and enabling the execution of AI models on devices with limited resources (Merenda et al., 2020).

Owing to these characteristics, Edge AI is used in various industrial applications, such as predictive maintenance to predict mechanical failures, quality control to automatically detect product defects, and process optimization to improve product efficiency through real-time decisions.

An example of the use of Edge AI is its integration with Digital Twins. A Digital Twin is a virtual copy of a machine or physical process that is continuously updated with data collected by sensors. Edge AI allows the processing of that data and enables Digital Twins to monitor systems in real time, predict possible failures, and optimize industrial processes (Rahman et al., 2025).

4. The future of AI ethics, scalability, and sustainability in a rapidly evolving landscape

The AI ethics guidelines are usually based on the technical aspects of AI, such as algorithms and their functionalities, but they rarely focus on the business practices or the economic contexts in which they are implemented. The traditional ethical principles are fairness, transparency, sustainability, and accountability. In the guidelines, fairness and accountability were the most frequently mentioned. In particular, the guidelines emphasise the need to decrease algorithm biases and promote diversity in development teams. In addition, the guidelines highlight the necessity of introducing control mechanisms in organizations through regulations, public supervision, and independent authorities. Transparency and sustainability, even though they are not often mentioned in guidelines, are extremely important: transparency is useful not only to explain the technical functionalities but also the business practices, decision-making processes, and data collection and storage (Attard-Frost et al., 2022)

As previously mentioned, the last traditional ethics aspect that the guidelines mention is sustainability, which is based on numerous aspects.

Currently, to be sustainable, companies must consider social, economic, environmental, and organizational aspects and overcome several challenges, such as incorporating sustainability in daily decisions, documenting their impact adequately, understanding that the dimensions of sustainability are linked to each other, finding a balance between risks and benefits, and collaborating with the institution to find a common trade-off. To address all these challenges, a good solution could be to integrate an instrument that helps organizations to evaluate the sustainability of the AI systems using appropriate criteria and indicators (Rohde et al., 2024).

Moving on to scalability, it can be stated that AI presents several challenges: delays in data processing; data integration; balancing resources, monitoring public and private

cloud environments, and coordinating all platforms; and privacy and security. To overcome these issues, numerous solutions can be applied, such as Federated Learning, which allows training distributed models without moving data to improve privacy, decrease data traffic, and increase scalability; Quantum Computing, which can handle complex models, improve resource prediction, and increase processing speed; serverless computing and AutoML, which allow automatic resource allocation, cost reduction, automation of the life cycle of the models, and scaling of the systems based on demand; and Edge Computing, which reduces latency, distributes the workload between the edge and the cloud, improves scalability, and reduces processing costs (Jamaludin et al., 2024).

5. Conclusion

In conclusion, from this discussion, it can be stated that the future of AI does not depend only on developing powerful models but also on ensuring that these systems are ethical, scalable, and sustainable. To achieve these goals, it is important to strengthen collaboration between institutions, industries, and researchers and adopt advanced technologies capable of balancing performance, security, and environmental impact.

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